

Skewness, Regression and Correlation

Module 2

Skewness

It is the *degree of distortion* from the symmetrical bell curve .
It measures the lack of symmetry in data distribution.

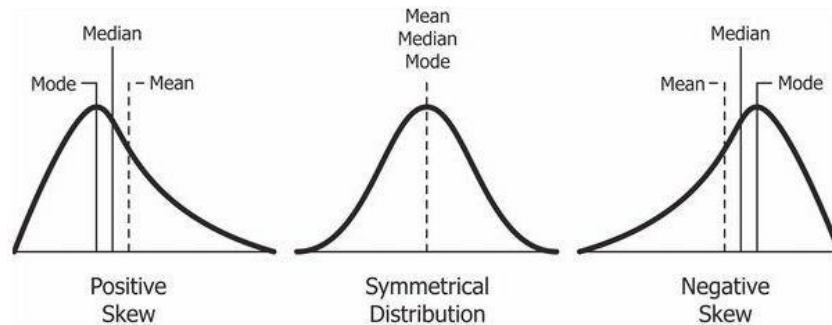
It differentiates extreme values in one versus the other tail. A symmetrical distribution will have a skewness of 0.

Skewness

There are two types of Skewness: Positive and Negative.

Positive Skewness means when the tail on the right side of the distribution is longer or fatter. The mean and median will be greater than the mode.

Negative Skewness is when the tail of the left side of the distribution is longer or fatter than the tail on the right side. The mean and median will be less than the mode.



Skewness

- If the skewness is between -0.5 and 0.5, the data are fairly symmetrical.
- If the skewness is between -1 and -0.5(negatively skewed) or between 0.5 and 1(positively skewed), the data are moderately skewed.
- If the skewness is less than -1(negatively skewed) or greater than 1(positively skewed), the data are highly skewed.

Karl Pearson's coefficient of skewness

$$\text{K.P coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D}}$$

Note: if there is more than one mode or no mode in distribution then it is defined by-

$$\text{K.P coefficient of skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{S.D}}$$

PROBLEM

Compute K.P coefficient of skewness for the following data

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	15	15	23	22	25	10	5	10

Bowley's Coefficient of Skewness

Bowley's Coefficient of Skewness

$$= \frac{(Q_1 - 2Q_2 + Q_3)}{Q_3 - Q_1}$$

Q_1, Q_2, Q_3 are quartiles.

PROBLEM

Compute Bowley's coefficient of skewness for the following data

C.I	30-35	35-40	40-45	45-50	50-55	55-60
Frequency	5	10	30	35	15	5

Skewness in real life

<https://youtu.be/XSSRrVMOqIQ>

Correlation and Regression

Correlation Analysis

Correlation analysis is applied in quantifying the association between two continuous variables, for example, an dependent and independent variable or among two independent variables.

Regression Analysis

Regression analysis refers to assessing the relationship between the outcome variable and one or more variables. The outcome variable is known as the dependent or response variable and the risk elements, and cofounders are known as predictors or independent variables. The dependent variable is shown by “y” and independent variables are shown by “x” in regression analysis.

The sample of a correlation coefficient is estimated in the correlation analysis. It ranges between -1 and +1, denoted by r and quantifies the strength and direction of the linear association among two variables. The correlation among two variables can either be positive, i.e. a higher level of one variable is related to a higher level of another or negative, i.e. a higher level of one variable is related to a lower level of the other.

The sign of the coefficient of correlation shows the direction of the association. The magnitude of the coefficient shows the strength of the association.

For example, a correlation of $r = 0.8$ indicates a positive and strong association among two variables, while a correlation of $r = -0.3$ shows a negative and weak association. A correlation near to zero shows the non-existence of linear association among two continuous variables

Karl Pearson's Coefficient of Correlation

Pearson's Correlation Coefficient is a linear **correlation coefficient** that returns a value of between -1 and +1. , -1 means there is a strong negative **correlation** and +1 means that there is a strong positive **correlation**. A 0 means that there is no **correlation** (this is also called zero **correlation**).

e.g the relationship between the age of a consumer and the color of shirt they might purchase or the level of education of a consumer and the delivery mechanism.

The coefficient of correlation r_{xy} between two variables x and y , for the bivariate dataset (x_i, y_i) where $i = 1, 2, 3, \dots, N$; is given by –

$$r_{(x,y)} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$\text{cov}(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

$\text{cov}(x,y)$: the covariance between x and y and σ_x and σ_y are the standard deviations of the distributions x and y .

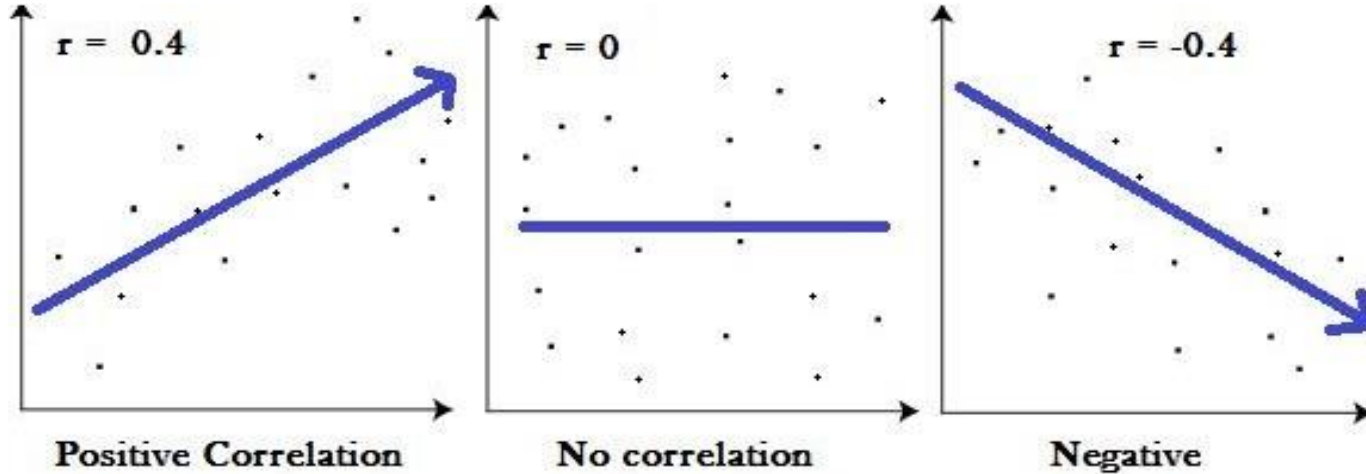
Karl Pearson's Coefficient of Correlation

A Single Formula for Discrete Datasets –

$$r_{xy} = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{N\sum x_i^2 - (\sum x_i)^2} \sqrt{N\sum y_i^2 - (\sum y_i)^2}}$$

r is unit-less. Thus, we may use it to compare association between totally different bivariate distributions as well.

Graphs showing a correlation of -1, 0 and +1



Scatter diagram

- 1 indicates a strong positive relationship.
- -1 indicates a strong negative relationship.
- A result of zero indicates no relationship at all.

Example:

Example question: Find the value of the correlation coefficient from the following table:

SUBJECT	AGE X	GLUCOSE LEVEL Y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

SUBJECT	AGE X	GLUCOSE LEVEL Y	XY	x^2	y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

From our table:

- $\sum x = 247$
- $\sum y = 486$
- $\sum xy = 20,485$
- $\sum x^2 = 11,409$
- $\sum y^2 = 40,022$
- n is the **sample size**, in our case = 6

The correlation coefficient =

- $6(20,485) - (247 \times 486) / [\sqrt{[6(11,409) - (247^2)] \times [6(40,022) - 486^2]}]$
= 0.5298

The **range** of the correlation coefficient is from -1 to 1. Our result is 0.5298 or 52.98%, which means the variables have a moderate positive correlation.

PROBLEM

Calculate Karl Pearson's coefficient of correlation between x and y for the following data and interpret the result:

$(1,6), (2,5), (3,7), (4,9), (5,8), (6,10), (7,11), (8,13), (9,12)$

ANSWER: 0.95

Spearman Rank Correlation Coefficient

The Spearman rank correlation coefficient, r_s , is the nonparametric version of the Pearson correlation coefficient. Your data must be ordinal, interval or ratio.

Spearman's returns a value from -1 to 1, where:

+1 = a perfect positive correlation between ranks

-1 = a perfect negative correlation between ranks

0 = no correlation between ranks.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

The scores for nine students in physics and math are as follows:

- Physics: 35, 23, 47, 17, 10, 43, 9, 6, 28
- Mathematics: 30, 33, 45, 23, 8, 49, 12, 4, 31

Compute the student's ranks in the two subjects and compute the Spearman rank correlation

Step 1: Find the ranks for each individual subject. Find the ranks by ordering the scores from greatest to smallest; assign the rank 1 to the highest score, 2 to the next highest and so on:.

Physics	Rank	Math	Rank
35	3	30	5
23	5	33	3
47	1	45	2
17	6	23	6
10	7	8	8
43	2	49	1
9	8	12	7
6	9	4	9
28	4	31	4

Step 2: Add a third column, d, to your data. The d is the difference between ranks. For example, the first student's physics rank is 3 and math rank is 5, so the difference is 3 points. In a fourth column, square your d values.

Physics	Rank	Math	Rank	d	d squared
35	3	30	5	2	4
23	5	33	3	2	4
47	1	45	2	1	1
17	6	23	6	0	0
10	7	8	8	1	1
43	2	49	1	1	1
9	8	12	7	1	1
6	9	4	9	0	0
28	4	31	4	0	0

Step 3: Sum (add up) all of your d-squared values. $4 + 4 + 1 + 0 + 1 + 1 + 1 + 0 + 0 = 12$. You'll need this for the formula (the $\sum d^2$ is just "the sum of d-squared values").

Step 4: Insert the values into the formula. These ranks are not tied, so use the first formula:

$$= 1 - (6*12)/(9(81-1))$$

$$= 1 - 72/720$$

$$= 1-0.1$$

$$= 0.9$$

The Spearman Rank Correlation for this set of data is 0.9.

Spearman Rank Correlation: What to do with Tied Ranks

Tied ranks are where two items in a column have the same rank. They are given the mean rank.

E.g

Xi	Rank(Xi)	Yi	Rank(Yi)	d_i^2
20	8	18	8	0
23	6.5	22	7	0.25
23	6.5	24	6	0.25
25	5	29	5	0
27	3.5	33	4	0.25
27	3.5	36	2	2.25
32	2	36	2	0
45	1	36	2	1
Total				4

To calculate ranks

For Xi

Rank 3,4 are tied hence
 $(3+4)/2 = 7/2 = 3.5$

Rank 6,7 are tied hence
 $(6+7)/2 = 13/2 = 6.5$

For Yi

Rank 1,2,3 are tied hence
 $(1+2+3)/3 = 6/3 = 2$

Correction in formula

m -> number of times rank repeated
 $(m^3 - m)/12$ is added to $\sum d_i^2$

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

Spearman Rank Correlation: What to do with Tied Ranks

Rank 3.5 is repeated $m=2$ times, hence correction $(m^3-m)/12=(2^3-2)/12=0.5$

Rank 6.5 is repeated $m=2$ times, hence correction $(m^3-m)/12=(2^3-2)/12=0.5$

Rank 2 is repeated $m=3$ times, hence correction $(m^3-m)/12=(3^3-3)/12=2$

Correction $\sum d_i^2 = 4 + 0.5 + 0.5 + 2 = 7$

Hence ρ is given by -

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

$$\rho = 1 - \frac{6 \times 7}{8(8^2 - 1)} = 0.92$$

PROBLEM

1. Two referees in a flower beauty competition rank the 10 types of flowers as follows:

Referee A	1	6	5	10	3	2	4	9	7	8
Referee B	6	4	9	8	1	2	3	10	5	7

2. Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in the Commerce and Mathematics.

Marks in Commerce	15	20	28	12	40	60	20	80
Marks in Mathematics	40	30	50	30	20	10	30	60

Linear Regression

Linear regression attempts to model the relationship between two variables by fitting a **linear** equation to observed data. One variable is considered to be an explanatory variable, and the other is considered to be a dependent variable.

When a **correlation coefficient** shows that data is likely to be able to predict future outcomes and a scatter plot of the data appears to form a straight line, you can use simple linear regression to find a predictive function.

Lines of regression - We measure the deviations of the points vertically and horizontally along y axis and x axis. Thus we have two lines of regression.

- a. Lines of regression Y on X
- b. Lines of regression X on Y

Least Square Method

It is the mathematical tool useful to obtain lines of regression of X on Y and Y on X.

i) Lines of regression of Y on X

$$Y = a + bX$$

ii) Lines of regression of X on Y

$$X = a + bY$$

Coefficients of regression

i) Coefficient of regression of Y on X

It is the slope of the line of regression of Y on X i.e **b** from the equation $y=a +bx$, it is given by

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

r = coefficient of correlation

σ_x = standard deviation of x

σ_y = standard deviation of y

$$b_{yx} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

line of regression of Y on X is $(y - \bar{y}) = b_{yx}(x - \bar{x})$

Coefficients of regression

ii) Coefficients of regression of X on Y

It is the slope of the line of regression of X on Y i.e **b** from the equation $x=a +by$, it is given by

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

r = coefficient of correlation

σ_x = standard deviation of x

σ_y = standard deviation of y

$$b_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(y_i - \bar{y})^2}$$

line of regression of X on Y is $(x - \bar{x}) = b_{xy}(y - \bar{y})$

PROBLEM

1. Find the equation of line of regression of y on x , line of regression of x on y and estimate the most probable value of y when $x=8$.

X	3	4	5	6	4	5	6	7
Y	3	5	3	2	3	4	6	6

Solution

x	y	$(x - \bar{x})$	$(x - \bar{x})^2$	$(y - \bar{y})$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
3	3	-2	4	-1	1	2
4	5	-1	1	1	1	-1
5	3	0	0	-1	1	0
6	2	1	1	-2	4	-2
4	3	-1	1	-1	1	1
5	4	0	0	0	0	0
6	6	1	1	2	4	2
7	6	2	4	2	4	4
Total:40	32		12		16	6

$$\begin{aligned}\bar{x} &= \sum x / n \\ &= 40/8=5\end{aligned}$$

$$\begin{aligned}\bar{y} &= \sum y / n \\ &= 32/8=4\end{aligned}$$

Coefficient of regression of Y on X is

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$
$$= 6/12 = 0.5$$

Line of regression of y on x is $(y - \bar{y}) = b_{yx}(x - \bar{x})$

$$y - 4 = 0.5(x - 5)$$

$$y = 0.5x + 1.5$$

When $x=8$

$$y = (0.5)(8) + 1.5 = 5.5 \approx 6$$

Line of regression for x on y

$$b_{xy} = 6/16 = 0.375$$

$$(x - \bar{x}) = b_{xy}(y - \bar{y})$$

$$x = 0.375y + 3.5$$

PROBLEM

1. Find the equation of lines of regression and estimate the most probable value of y when $x=15.5$.

X	10	12	13	16	17	20	25
Y	19	22	24	27	29	33	37

Non- Linear Regression

Non- Linear Regression

Nonlinear regression is a form of **regression** analysis in which observational data are modeled by a function which is a **nonlinear** combination of the **model** parameters and depends on one or more independent variables. The data are fitted by a method of successive approximations.

Nonlinear models are more complicated than linear models to develop because the function is created through a series of approximations (iterations) that may stem from trial-and-error.

- Quadratic regression
- Cubic regression

Quadratic Regression

Quadratic regression is finding the best fit equation for a set of data shaped like a parabola.

The first step in regression is to **make a scatter plot**. If your scatter plot is in a “U” shape, either concave up (like the letter U) or concave down ($\cap$), you’re probably looking at some type of quadratic equation as the best fit for your data. A quadratic doesn’t have to be a full “U” shape; you can have part of a it (say, a quarter or 3/4).

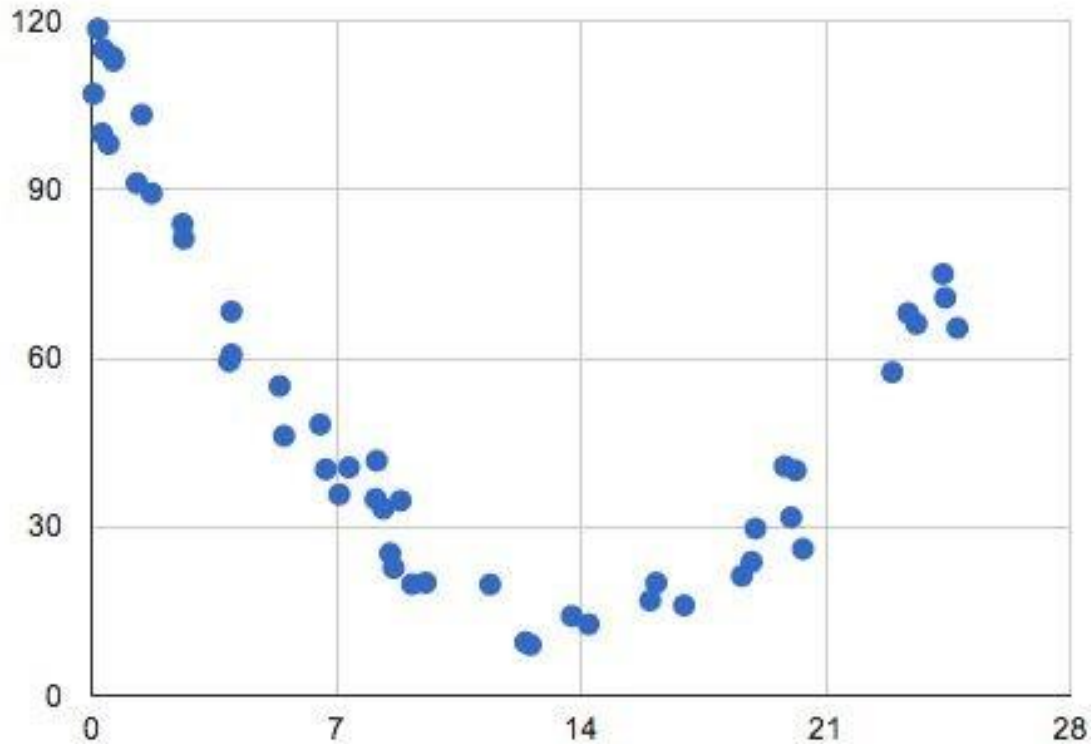
Quadratic regression is an **extension of simple linear regression**. While linear regression can be performed with as few as two points (i.e. enough points to draw a straight line), quadratic regression come with the disadvantage that it requires more data points to be certain your data falls into the “U” shape.

Quadratic regression is a way to model a relationship between two sets of variables. The result is a regression equation that can be used to make predictions about the data. The equation has the form:

$$y = ax^2 + bx + c,$$

where $a \neq 0$.

Data points that suggest quadratic regression would be a good fit.



Quadratic Regression

In order to find the quadratic regression by hand, you have to solve the following **system of equations**. This set of equations is sometimes called *normal equations*.

$$a\sum x_i^4 + b\sum x_i^3 + c\sum x_i^2 = \sum x_i^2 y_i$$

$$a\sum x_i^3 + b\sum x_i^2 + c\sum x_i = \sum x_i y_i$$

$$a\sum x_i^2 + b\sum x_i + cn_i = \sum y_i$$

PROBLEM

1. Find the quadratic regression equation for the following set of data

x: 1, 3, 5, 7, 9

y: 32.5, 37.3, 36.4, 32.4, 28.5

SOLUTION

Make a table . Input your x-values in the first column and your y-values in the second column. Add 5 more columns labeled x^2 , x^3 , x^4 , xy , and $x^2 * y$:

x	y	x^2	x^3	x^4	xy	$x^2 * y$
1	32.5					
3	37.3					
5	36.4					
7	32.4					
9	28.5					

Calculate each column. For example, the x^2 column is simply the squares of the first column; the last column is the third column multiplied by the second column (the y -values)

	x	y	x^2	x^3	x^4	xy	$x^2 * y$
	1	32.5	1	1	1	32.5	32.5
	3	37.3	9	27	81	111.9	335.7
	5	36.4	25	125	625	182	910
	7	32.4	49	343	2401	226.8	1587.6
	9	28.5	81	729	6561	256.5	2308.5
TOTAL	25	167.1	165	1225	9669	809.7	5174.3

Use the last row (the summations) to fill in the values. All you're doing is transferring the numbers to the normal equation (n is the number of items in the set, which is 5 in our QUESTION)

$$a \sum x_i^4 + b \sum x_i^3 + c \sum x_i^2 = \sum x_i^2 y_i$$
$$9669a + 1225b + 165c = 5174.3$$

$$a \sum x_i^3 + b \sum x_i^2 + c \sum x_i = \sum x_i y_i$$
$$1225a + 165b + 25c = 809.7$$

$$a \sum x_i^2 + b \sum x_i + cn_i = \sum y_i$$
$$165a + 25b + 5c = 167.1$$

Solve the system of equations to get the values.

$$a = -0.3660714$$

$$b = 3.015714$$

$$c = 30.42179$$

Insert the values into the quadratic equation (I'm rounding to 3 decimal places).

$$y = ax^2 + bx + c$$

$$y = -0.366x^2 + 3.016x + 30.422$$

Find the quadratic regression with a Calculator

<https://www.youtube.com/watch?v=FWsJjyWrcKs>

https://www.youtube.com/watch?v=1b4I-N1vH_0